

Solution Requirements

Date	29 September 2026
Team ID	6ab3e6756603573ccfb3b891
Project Name	FitBuddy - AI Fitness Plan Generator Using Gemini Models
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

Functional requirements describe what FitBuddy does; non-functional requirements describe the operating qualities expected from the current educational implementation.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Current version does not include a user login system; the User ID field identifies records. Authentication is a future enhancement.	Medium
2	Authorization	Current version provides an admin-style users page but does not implement role-based authorization.	Medium
3	External interfaces	Integrate with Google Gemini through the Google GenAI SDK for workout, nutrition and plan-update generation.	High
4	Transactions	Process form submission, save user data, save generated plan, collect feedback and save updated plan.	High
5	Reporting	Provide an admin page that lists saved users and their latest stored plans.	Medium
6	Business rules	Accept age 13-100, weight >20 and <400 kg, fixed goals (weight loss, muscle gain, general wellness) and intensity (low, medium, high).	High
7	Compliance	Avoid medical diagnosis or extreme dieting in generated prompts; protect API keys via .env and exclude them from Git.	High
8	Other	Provide a health endpoint and FastAPI interactive API documentation.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Home, health and documentation endpoints should return promptly during normal local use.	Qualitative smoke check; no formal load benchmark claimed.
2	Scalability	Separate routing, AI and database modules allow later migration to a hosted database and scaled API service.	Architecture supports modular replacement.
3	Security & Data Privacy	Do not hard-code API keys; validate inputs and keep .env out of source control.	.env in .gitignore; Pydantic validation enabled.
4	Reliability & Availability	Return a controlled error page when generation fails.	HTTP error handling implemented in routes.
5	Usability & Accessibility	Responsive form and result pages with clear fields, labels and navigation.	Usable on common desktop/mobile browser sizes.
6	Other - Maintainability	Keep AI generation, routes, schemas, database and templates separated.	Module-based code layout.